

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Industry Problem Solving Task

Criteria	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	Total (10)
Weightage	20%	25%	25%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I KONGARA JOGESH(192525035) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:K.jogesh

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

CSA16 – DATA WAREHOUSING AND DATA MINING

1. Introduction

Clustering is an unsupervised data-mining technique that groups observations according to similarity. In industry, clustering can support customer segmentation, fraud detection, churn analysis, recommendation systems and traffic management. The quality of a clustering solution depends not only on the algorithm but also on feature selection, preprocessing, distance or similarity measures, validation, domain interpretation and deployment constraints.

For each problem in this assessment, the solution therefore follows an engineering workflow: understand the business problem, prepare the data, select a suitable clustering method, design an implementation pipeline, interpret the clusters, validate the result and connect the output to an operational decision.

2. Q1 – Supermarket Purchases: Customer Segmentation

Industry problem: Using visits per month, average spend and items bought, propose a clustering approach to segment customers and explain how it supports targeted marketing.

Customer	Visits / Month	Avg Spend (₹)	Items Bought
C1	10	2,000	15
C2	4	5,000	8
C3	12	1,800	18
C4	3	6,000	6
C5	11	2,100	16

Customer Segmentation: Visits vs Average Spend



Figure 1. Customer behaviour visualization from the supplied dataset.

2.1 Understanding of the industry problem

The objective is to convert raw purchase behaviour into meaningful customer groups that marketing teams can act upon. The three features capture frequency, monetary value and basket size. A useful

segmentation should be interpretable enough to drive different offers rather than simply producing mathematically distinct groups.

2.2 Engineering concept and algorithm

K-means is a suitable starting point because the dataset contains numerical attributes and the goal is to create behavioural segments. Before clustering, the variables should be standardized because average spend is measured in thousands of rupees while visits and item counts are much smaller. A practical workflow is: clean data → standardize features → select k using an elbow/silhouette analysis → run K-means → profile each cluster.

2.3 Evidence from the dataset

- C2 and C4 visit less frequently but have high average spending (₹5,000 and ₹6,000). They may represent high-value but low-frequency customers.
- C1, C3 and C5 visit frequently (10–12 visits) and buy 15–18 items, while their average spending is lower (₹1,800–₹2,100). They may represent frequent, regular shoppers.
- The contrast suggests that customer value cannot be determined from one feature alone; clustering combines frequency, monetary value and basket behaviour.

2.4 Solution design

- Preprocess: check missing/duplicate records and standardize the three numerical variables.
- Cluster: begin with k=2 or k=3 and compare candidate values using silhouette score and business interpretability.
- Profile: calculate the average feature value for each cluster and assign meaningful business labels such as High-Value/Low-Frequency and Frequent/Regular.
- Act: provide premium offers to high-value customers, loyalty rewards to frequent customers and re-engagement offers to low-activity groups.
- Monitor: compare conversion rate, average order value and repeat-purchase rate before and after targeted campaigns.

2.5 Tools and implementation

Python with pandas can be used for data preparation, scikit-learn for StandardScaler and KMeans, matplotlib for visualization and a notebook environment for reproducible analysis. In an industry deployment, the resulting cluster label can be stored in a customer table or CRM segmentation field.

2.6 Validation and analysis

Validation should combine silhouette score with domain review. A numerically strong cluster is not automatically useful if marketing staff cannot interpret it or if two clusters receive the same campaign. A/B testing can validate whether cluster-based campaigns improve response and revenue.

Conclusion: A standardized K-means workflow is appropriate for this small numerical dataset and can convert customer behaviour into actionable marketing segments.

3. Q2 – Credit Card Transactions: Unusual/Fraudulent Transaction Detection

Industry problem: Design a clustering-based solution using transaction amount, location and time to identify unusual or potentially fraudulent transactions.

Transaction	Amount (₹)	Location	Time (hrs)
T1	200	Chennai	10
T2	5,000	Delhi	2
T3	150	Chennai	11
T4	7,000	Mumbai	1
T5	180	Chennai	12

Transaction Amount vs Time

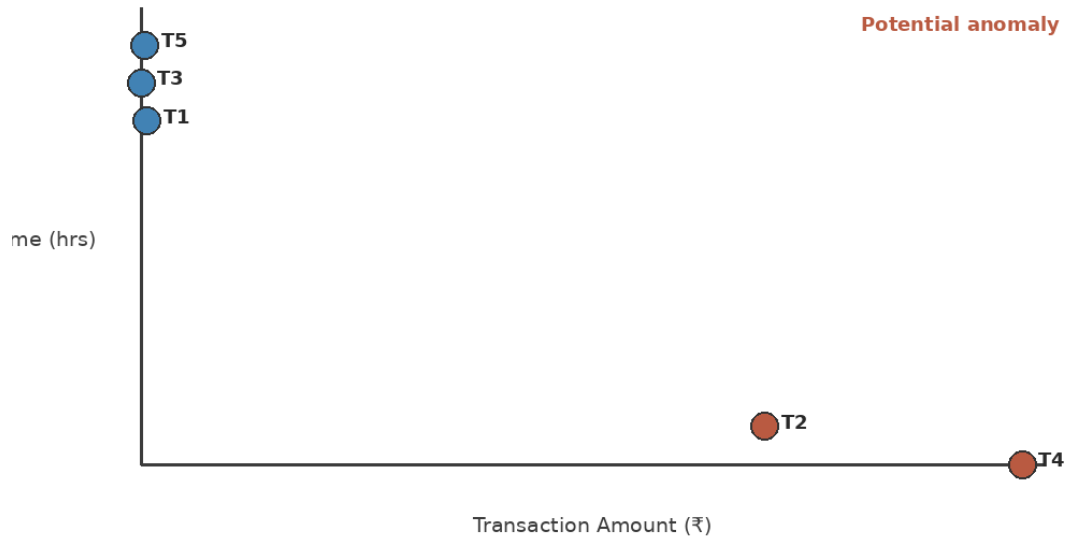


Figure 2. Transaction amount and time showing the two high-value transactions as potential anomalies.

3.1 Problem understanding

Fraud detection is an anomaly-detection problem. Legitimate transactions may form dense behavioural patterns, while unusual combinations of amount, location and time can appear as sparse points. Clustering can therefore be used as an unsupervised screening layer, especially when labelled fraud examples are limited.

3.2 Engineering concept

A density-based method such as DBSCAN is suitable because it can identify dense groups and label isolated observations as noise. K-means can also be used to identify distant points from centroids, but it forces every observation into a cluster and therefore requires an additional anomaly-score rule.

3.3 Evidence

- T1, T3 and T5 are low-value Chennai transactions occurring between 10 and 12 hours, forming a coherent behavioural pattern.
- T2 and T4 are much higher-value transactions (₹5,000 and ₹7,000) and occur at 2 and 1 hours in different locations.

- The difference in amount, time and location makes T2 and T4 candidates for investigation, although clustering alone cannot prove fraud.

3.4 Solution design

- Encode location appropriately and scale numerical features so transaction amount does not overwhelm time/location information.
- Run DBSCAN with parameters selected through neighbourhood-distance analysis.
- Treat noise points as alerts and compute an anomaly score using distance from the nearest dense cluster or local density.
- Add business rules such as impossible travel, unusual hour, unusually high amount and rapid location change.
- Send high-risk transactions for step-up authentication or manual review rather than automatically declaring them fraudulent.
- Continuously retrain/refresh clusters as spending patterns change.

3.5 Tools and validation

Python, pandas, scikit-learn and matplotlib can implement the pipeline. A production system can connect the anomaly score to a streaming transaction service. Validation should use precision, recall, false-positive rate and investigation outcomes when confirmed fraud labels become available.

Conclusion: Density-based clustering is a strong unsupervised screening method for this dataset because the normal Chennai pattern is dense while the high-value, unusual-time transactions stand apart. Human/business validation is essential before treating any alert as fraud.

4. Q3 – Telecom Usage: Churn-Prone User Identification

Industry problem: Explain how clustering can identify churn-prone users and improve retention strategies.

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Telecom Usage Segmentation

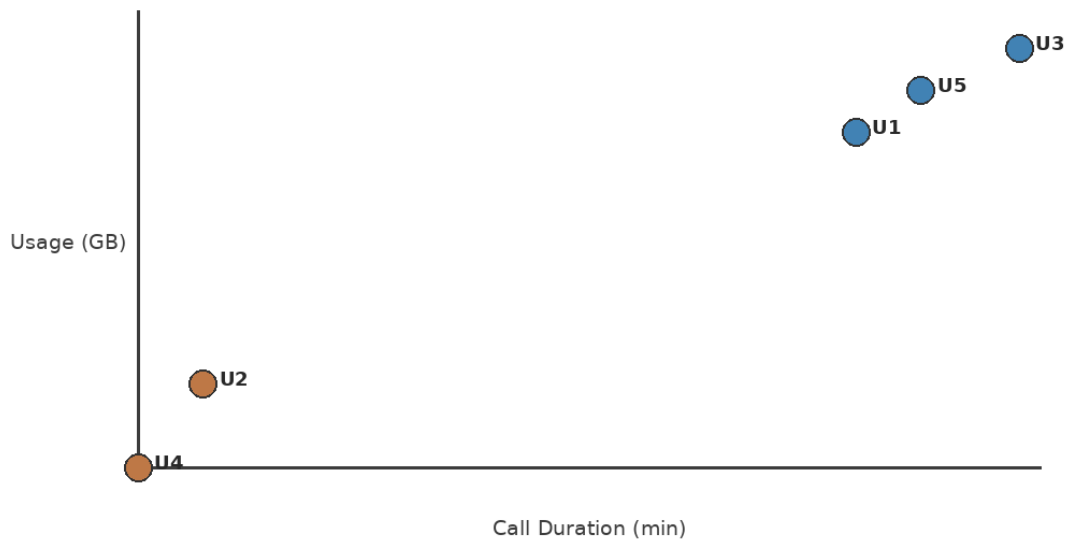


Figure 3. Telecom users grouped by call duration and data usage.

4.1 Problem understanding

Churn is the loss of customers to competitors or non-renewal. The dataset provides usage intensity and recharge value, which can be used to discover behavioural segments. A cluster is not automatically a churn label; it is a behavioural group that must be linked with historical churn outcomes.

4.2 Engineering concept

K-means or hierarchical clustering can group users after standardization. For example, users with high call duration, high data usage and high recharge can form an active-value segment, while low-usage/low-recharge users can form a low-engagement segment. If historical churn labels are later available, cluster-level churn rates can be calculated.

4.3 Evidence

- U3 and U5 have high call duration, high data usage and high recharge, indicating strong engagement.
- U2 and U4 have lower usage and recharge values, indicating a lower-engagement segment.
- U1 sits closer to the high-usage group but below U3/U5, illustrating why clustering can create more nuanced segments than a single threshold.

4.4 Solution design and retention strategy

- Standardize call duration, data usage and recharge.
- Use K-means with candidate k values and validate with silhouette score.
- Profile each cluster and calculate historical churn rate if labelled records are available.
- Prioritize low-engagement/high-churn clusters for retention campaigns such as tailored data packs, service-quality follow-up or loyalty benefits.

- Use high-engagement clusters for upselling and loyalty programmes rather than unnecessary discounts.
- Track churn rate, retention rate and campaign conversion by cluster.

4.5 Validation

The strongest validation is to compare observed churn rates across clusters on historical data and then test whether cluster-targeted interventions reduce churn. Cluster stability should also be checked over multiple time periods.

Conclusion: Clustering can expose behavioural segments and make retention campaigns more targeted, but churn risk should be validated using actual historical churn outcomes.

5. Q4 – E-commerce Behaviour: User Grouping for Recommendation Systems

Industry problem: Suggest a clustering technique to group users and explain how it improves recommendation systems.

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

E-commerce Engagement Segmentation

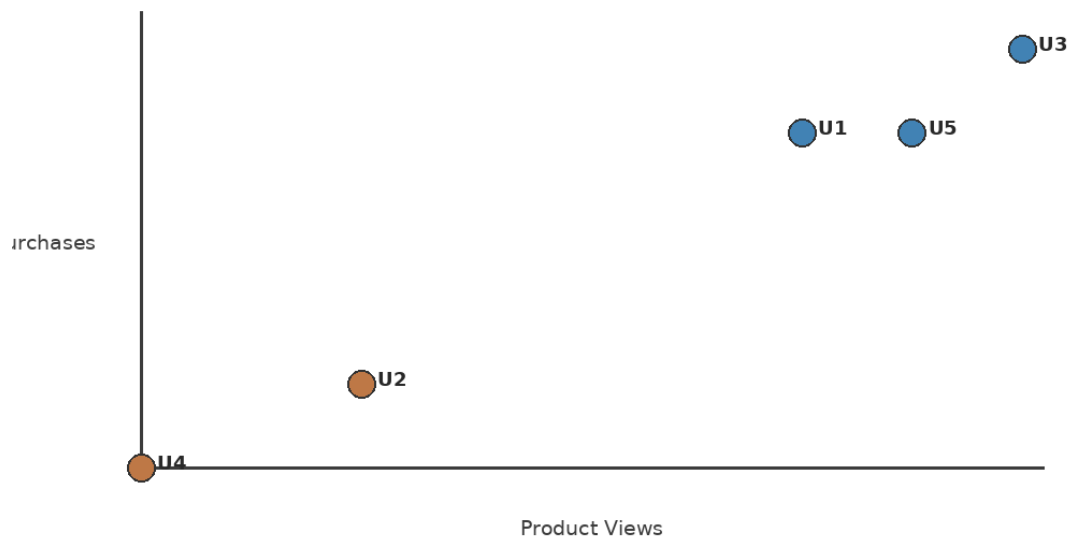


Figure 4. E-commerce engagement pattern from views and purchases.

5.1 Problem understanding

Recommendation systems need to estimate what a user is likely to engage with. Clustering can create behavioural cohorts so that users with similar browsing and purchase patterns can receive relevant

product recommendations. This is particularly useful as a cold-start or cohort-based strategy when individual user history is limited.

5.2 Engineering concept

K-means is suitable for this small numerical dataset after scaling. Users can be grouped according to views, clicks and purchases. A more advanced system can combine cluster-based recommendations with item similarity, collaborative filtering or content-based ranking.

5.3 Evidence

- U3 has the highest engagement: 60 views, 25 clicks and 6 purchases.
- U1 and U5 are also highly engaged with 50/20/5 and 55/22/5 respectively.
- U4 is the least engaged with 20 views, 8 clicks and 1 purchase.
- The behavioural separation supports cohort-based recommendation strategies.

5.4 Solution design

- Standardize views, clicks and purchases.
- Cluster users into behavioural groups and calculate the dominant product categories/items within each group.
- Generate recommendations from the cluster's popular items, then refine ranking using each user's individual history.
- Use purchase conversion and click-through rate to evaluate recommendation quality.
- Refresh clusters periodically because user behaviour changes.

5.5 Tools and analysis

Python/pandas can prepare user-event data, scikit-learn can perform clustering, and matplotlib can visualize user segments. A production recommendation API can retrieve a user's cluster and combine cohort popularity with personalized ranking.

Conclusion: K-means-based user cohorts can improve recommendation relevance by grouping users with similar engagement behaviour, especially when combined with individual personalization.

6. Q5 – Traffic Data: Congestion-Zone Identification

Industry problem: Design a clustering-based solution to identify traffic congestion zones and explain how it helps traffic management.

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Traffic Congestion Pattern

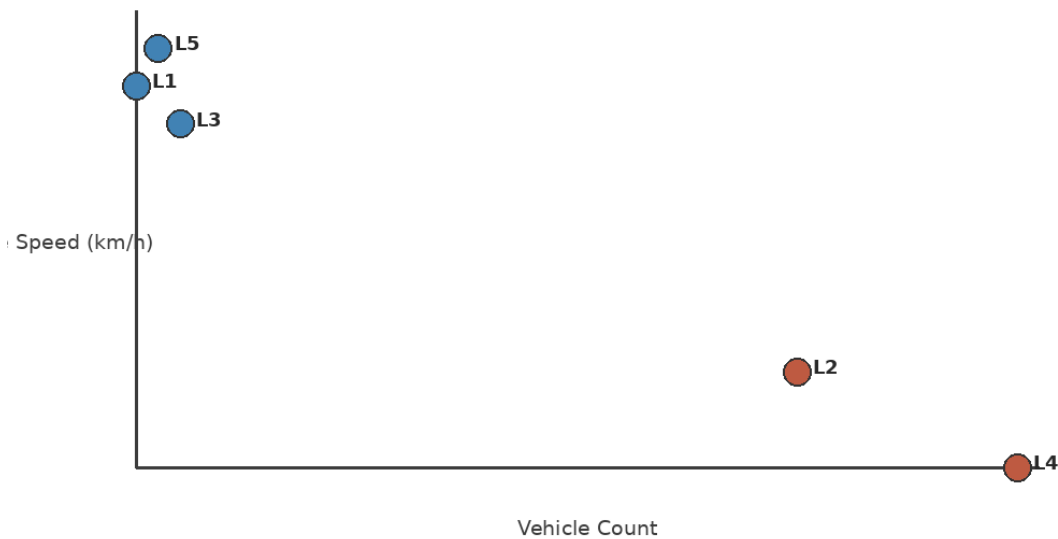


Figure 5. Vehicle count and average speed indicate high-congestion locations L2 and L4.

6.1 Problem understanding

Traffic management requires identifying locations where vehicle volume is high, speed is low and delay is large. Clustering can group locations with similar traffic conditions so that interventions can be prioritized.

6.2 Engineering concept

K-means can create traffic-condition clusters after standardizing vehicle count, average speed and delay. Because congestion can have irregular patterns and noise, density-based methods may also be considered when many locations are available. For this small dataset, K-means provides a simple interpretable starting point.

6.3 Evidence

- L2 has 500 vehicles, an average speed of 15 km/h and 25 minutes of delay.
- L4 is even more severe with 600 vehicles, 10 km/h average speed and 30 minutes of delay.
- L1, L3 and L5 have lower vehicle counts, higher speeds and lower delays, indicating a less congested group.
- The separation supports a high-congestion cluster and a lower-congestion cluster.

6.4 Solution design

- Collect traffic observations over multiple time windows rather than relying on one measurement.
- Clean missing or faulty sensor values and standardize the three traffic features.
- Run clustering and profile each cluster using average count, speed and delay.
- Map high-congestion cluster locations to a GIS/dashboard.

- Use results for signal timing, route diversion, public transport prioritization, incident response and road-capacity planning.
- Validate clusters using subsequent traffic observations and compare congestion before and after interventions.

6.5 Tools and validation

Python/pandas and scikit-learn can implement the clustering pipeline; matplotlib or a GIS/dashboard tool can visualize zones. Real-time deployments can connect traffic sensors or APIs to the pipeline. Validation should compare cluster labels with observed delay, average speed and congestion incidents.

Conclusion: Clustering provides a practical way to convert traffic measurements into operational congestion zones. L2 and L4 are clear high-priority candidates in the supplied data.

7. Rubric-Based Quality Alignment

The assessment rubric gives 20% to understanding the industry problem, 25% to application of engineering concepts, 25% to solution design, 15% to use of tools, 10% to analysis and 5% to professionalism. The following map shows how this submission targets Level 4.

Rubric Criterion	Weightage	Evidence in This Assessment
Understanding of Industry Problem	20%	Each question begins with the real-world objective, constraints and business meaning of the supplied data.
Application of Engineering Concepts	25%	Algorithm selection, preprocessing, feature scaling, clustering workflow, anomaly detection, validation and deployment concepts are explained.
Solution Design	25%	Each problem includes a practical pipeline from data preparation through clustering, interpretation, action and monitoring.
Use of Tools	15%	Python, pandas, scikit-learn, matplotlib, dashboards/GIS and production integration are identified where appropriate.
Analysis	10%	Evidence from the supplied values is interpreted, limitations are stated and validation/performance measures are proposed.
Professionalism	5%	Editable front pages, rubric page, structured headings, tables, figures, conclusions and consistent formatting are included.

8. Cross-Problem Comparison

Industry Problem	Suggested Method	Primary Output	Business Action
Customer segmentation	K-means	Customer clusters	Targeted marketing and loyalty offers
Fraud screening	DBSCAN / anomaly clustering	Normal vs unusual transaction groups	Investigation and step-up verification
Churn analysis	K-means / hierarchical	Usage-behaviour segments	Retention campaigns

Recommendation	K-means cohorts + recommender	User behaviour groups	Personalized/cohort recommendations
Traffic management	K-means / density clustering	Congestion zones	Signal, routing and capacity interventions

9. Overall Conclusion

The five industry scenarios demonstrate that clustering is most effective when algorithm choice is connected to the real-world objective. K-means is a strong baseline for numerical segmentation, DBSCAN is valuable for density and anomaly-oriented problems, and hierarchical approaches can support exploratory segmentation. Across all cases, preprocessing, feature scaling, validation and domain interpretation are essential. The proposed solutions transform the supplied datasets into actionable outputs: marketing segments, fraud alerts, churn-risk cohorts, recommendation groups and traffic congestion zones. These outputs can then be connected to measurable business outcomes such as campaign conversion, fraud investigation precision, retention rate, recommendation click-through and reduction in traffic delay.